INCH-POUND

MIL-DTL-1577E
8 January 2024
SUPERSEDING
MIL-P-1577D
30 September 1976

DETAIL SPECIFICATION SHEET

PANEL MARKER, RED CROSS, 8 FEET, 6 INCHES BY 19 FEET, 6 INCHES

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope: This specification covers one type of Red Cross panel marker, 8 feet 6 inches long by 19 feet 6 inches wide, made from laminated vinyl-nylon cloth.

2. APPLICABLE DOCUMENTS:

2.1 Government Document

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to extent specification herein. Unless otherwise specified, the issue of these documents will be these listed in the issue of the Department of Defense Index of specifications and standards and supplement thereto, cited in the solicitation.

COMMERCIAL ITEM DESCRIPTION

A-A-55308 – Cloth and Strip Laminated, Vinyl Nylon High Strength, Flexible
A-A-59963, Thread, Polyester

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-L-1709 – Line Tent

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 – Sampling Procedures and Tables for Inspection by Attributes
MIL-S NASM16491 – Grommet, Metallic
MIL-STD-129 – Marking for shipment and Storage.

Comments, suggestions, or questions on this document should be addressed to Department of the Army, The Institute of Heraldry, Technical and Production Division, 9325 Gunston Road, Building 1466, Room S112, Fort Belvoir, VA 22060-5579 or usarmy.belvoir.hqda.mbx.tioh-webmaster. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8345

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2.1.2 Other Government document, drawings and publications. The following other documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issue will be those in effect on the date of the solicitation.

DRAWING

THE INSTITUTE OF HERALDRY

5-1-380 – Panel Marker, Red Cross Laminated Vinyl-Nylon Cloth, 8’ 6” by 19’ 6” (See Figure 1)

(Copies of specifications, Standards and Drawings required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

(Figure 1 is a miniature reproduction of drawing 5-1-380 and is attached for information purpose only.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM-D5118/D5118M – Boxes, Shipping Fiberboard
ASTM-D-6193 – Stich, Seam and Stitching

(Copies for these documents are available online at https://www.astm.org)

THE COLOR ASSOCIATION OF THE UNITED STATES
AMERICAN AND EFIRD, LLC
DoD Standard Shades for Heraldic Yarns, 2012 Edition

(Copies for these documents are available online at www.salesRA@amefird.com)

(Nongovernment standards and other publications are normally available from the organizations which prepare, or which distribute the documents. These documents also may be available in or through libraries or other informational service.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheet or MS standards) the text of this specification will take precedence. Nothing in this specification, however, will supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Standard sample. Unless otherwise specified (see 6.2), the contracting officer shall furnish standard samples of the laminated vinyl-nylon cloth (both red and white), the thread and marking required.

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3.2 First article. When otherwise specified (see 6.2), before production is commenced, a sample of the finish panel marker covered by this specification 60-63 be submitted or made available to the contracting officer or authorized representative for inspection and testing as specified in 4.2. The approval of the first article authorizes the commencement of production but does not relieve the contactor of responsibility for compliance with all applicable provisions of this specification. The first article shall be manufactured by the contractor in the same facilities to be used for the manufacturing of the production items.

3.3 Materials. Materials shall conform to the referenced specifications and the requirements specified herein.

3.3.1 Cloth Laminated, Vinyl-Nylon. The laminated vinyl-nylon cloth for the panel marker shall conform to type ll class 2, form of A-A55308. The coated cloth for the cross portion of the marker shall be red shade 22136. The field portion of the marker shall be white shade 27875.

3.3.2 Slip Tent. The tent slips shall conform type II of MIL-DTL-1734

3.3.3 Thread Polyester. The polyester thread shall be natural (undyed) conforming to type I of MIL-DTL-32072. The thread sizes shall be as follows:

For all stitching, except stitching indicated by “Z” on drawing …. Size F for needle and bobbin.

For all stitching indicated by “Z” on drawing ……Size F for needle and size E for loopier.

3.3.4 Tent Lines. The tent lines for the marker shall to MIL-L-1709. The types and class shall be as follows:

For corner lines ……Type XXIV, Class A

For guy lines …… Type XXXVIII, Class A

3.3.5 Grommet. The grommets shall conform to type III, class 3 of NASM16491.

3.4 Construction. The panel marker shall be constructed in accordance with the details and dimensions of drawing 5-1-380 and the requirements specified herein. The panel marker shall be fabricated from vinyl-nylon laminated cloth specified in 3.3.1. The grommets specified in 3.3.5, shall be set by punching a hole through the material which is smaller than the outside diameter of the grommet barrels. Each of the grommet barrels shall be forced through the holes and securely clinched with-out cutting the material.

3.4.1 Stitching. All stitching shall be in accordance with ASTM-D6193, using thread specified in 3.3.3. For all stitching except stitching indicated by a “Z” on the drawing, a type 301 stitch shall be used with 5 to 7 stitches to the inch. For all stitching indicated by “Z” on the drawing a type 401 stitch shall be used with 5 to 7 stitches to the inch. When a 401 stitch is used, the chain portion of the stitching shall not appear on the face side of the marker. All thread ends shall be trimmed.

3.4.1.1 Type 301 Stitching. When a type 301 stitch is used, thread breaks shall be backstitched not less than one inch at each break. Except where ends are turned under in a hem or held down by other stitching, stitching ends shall be backstitched 1/2 inch to prevent raveling. Thread tension shall be maintained so that there will be no loose

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stitching and so that the lock will be imbedded in the center of the material sewn.

3.4.1.2 Type 401 Stitching. When type 401 stitch is used, thread breaks shall be overstitched not less 1-1/2 inches at each break. The thread tension shall be maintained so that there will be no loose stitching.

3.5 Repairs. Repairs whenever necessary and when authorized by the contracting officer, shall be effected by securely cementing or sewing a patch of the same color material to the damaged area.

3.6 Marking. The letter “US”, identification marking, and any marking required by applicable drawing shall be imprinted or stenciled in the locations shown on the drawing. When tested as specified in 4.4, all marking shall show fastness to weathering equal to or better than the standard sample (See 3.1). If no standard sample is available, the marking shall show good fastness to weathering. The identification marking shall be as follows:

APPROVED NOMENCLATURE
STOCK NUMBER
NUMBER AND DATE OF CONTRACT

3.7 Workmanship. The panel marker shall be clean, well made and shall meet the acceptable quality level established by this specification.

4 VERIFICATIONS

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows**:**

a**.** First article Inspection (see 4.2)
b**.** Quality Conformance inspection (see 4.3)

4.2 First Article inspection. Inspection and testing of the first article shall be made on finished panel marker for all provision of this specification applicable to the end item examination and tests.

4.3 Quality conformance inspection. Inspection shall be in accordance with the provision set forth in MIL-STD-1916, except where otherwise indicated.

4.3.1 Inspection of components.

4.3.1.1 Testing of components. In addition to Verification of the subsidiary specifications and standards, inspection shall be performed on samples of identification marking for the characteristic indicated in table I. The inspection level shall be S-I and the AQL shall be 6.5 defects per hundred units. Requirements shall be applicable to the individual unit.

TABLE I Component Testing,

Characteristic	Rqm’t para	Test Method	#Determ per S/U	Results Reported as
Fastness of marking	3.6	4.4	1	pass or fail

4.3.2 In-process Inspection. Inspection of the subassemblies shall be made at any point of the manufacturing process to determine whether operations or assemblies are accomplished as specified on the drawing. The government reserves the right to exclude from consideration for acceptance any assembly in which In-process inspection has

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indicated nonconformance.

4.3.3 End item inspection.

4.3.3.1 Visual examination. Defect found during the examination shall be classified in accordance with 4.3.3.1.1 and 4.3.3.1.2. Lots shall be in accordance with 4.3.3.1.3. The sample unit for this examination shall be one completely fabricated panel marker.

4.3.3.1.1 Examination for defects in material, design, construction, and workmanship.

Examine	CLASSIFICATION Defect	Major	Minor
Fabric	Any hole, cut or tear	X	-
	Any fabric defects:		
	-seriously affecting serviceability.....	X	-
	-affecting serviceability, but not seriously	-	X
Hardware	Broken or malformed; Finish omitted; Corroded areas; Burrs or sharp edges which may damage the item or Cause injury in handling	X	-
Grommets	Size, class, or type not specified:		
	-seriously affecting serviceability	X	-
	-affecting serviceability, but not seriously	-	X
	Clinched excessively tight, cutting fabric, or insecurely clinched, i.e., grommet or washer becomes disengaged	X	-
	End grommet set with rim of grommet extending beyond edge of hem	X	-
Guy and corner Lines	Eye or cut splice not as specified; free ends not finished as specified: less then specified length by more than 2 inches	-	X
Seams and Stitching	Wrong seam or stitch type	X	-
	Stitching margin or stitching gage not as specified, affecting serviceability	-	X
	Stich tension; loose, resulting in a loosely exposed top or bobbin thread; tight, resulting in excessive tightness or puckering of fabric on seams.....	X	-
Open seam	on stitch type 301:		
	-more than 1/2 inch, but not more than 1 inch	-	X
	-more than 1 inch	X	-
	On stitch type 401, any open seam except runoff	X	-
	Runoffs on stitch type 401:		
	-more than 1/2 inch but not more than 1 inch	-	X
	-more than 1 inch	X	-
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped stitches or runoffs occur. On double stitched seams, a seam shall be considered open when either one or both lines of stitching are open.		
Raw edge	When securely caught in stitching:		
	-more than 1 inch, but not more than 2 inches	-	X

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	-more than 2 inches	X	-
	NOTE: A raw edge not securely caught in stitching shall be classified as open seam.		
Runoffs	(see open seams)		
Stitches per Inch	One stitch less than minimum specified	-	X
	Two or more stitches less than minimum	X	-
	One stitch in excess of maximum specified	-	X
	Two or more stitches in excess of maximum specified.....	X	-
Thread breaks	Backstitched less than one inch on stitch type 301; overstitched less than 1-1/2 inches on stitch type 401	-	X
	NOTE: Thread breaks not backstitched or not overstitched shall be classified as open seam.		
Stitching	Backstitched less than 1/2 inch (except where ends are held down by other stitching or turned under in a hem)	-	X
	Not backstitched	X	-
Components	Chain portion of stitching shows on outside of the panel marker	-	X
	Any component part or required operation omitted (unless otherwise classified herein)	X	-
	Needle chews expected to develop into a hole	X	-
	Seams badly pleated or puckered	-	X
Repairs	Not as specified, or not authorized by the contracting officer.	X	-
Hems	Twisted or pleated	-	X
	Less than specified width:		
	-by more than 1/4 inch, but not more than 1/2 inch	-	X
	-by more than 1/2 inch	X	-
	Not constructed as specified in section “C-C” on dwg	-	X
Reinforcements	Misplaced:		
	-seriously affecting serviceability	X	-
	-affecting serviceability, but not seriously	-	X
	Finished dimensions less than specified by more than 1/2 inch	-	X
	Improperly applied, causing excessive fullness on reinforcements or reinforced parts	-	X
	Not stitched to reinforced parts on all sides or not caught in seams or stitching of reinforced parts	-	X
	Center piece set with center line off-center of marker by more than one inch	X	-
	Reinforcement “A” set with center off-center of center piece by more than 1/2 inch	-	X
Red Cross Panel	Vertical or horizontal center line of Red Cross off-center piece of marker halves:		
	-by more than 1/2 inch, but not more than one inch	-	X
	-by more than one inch	X	-
	Any part of the marker, except Red Cross, not white	X	-
Marking	Omitted, incorrect, illegible, or misplaced	-	X
Cleanness	Noticeable grease or oil stain; thread ends not trimmed		

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Throughout	-	X

4.3.3.1.2 Dimensional defect. Any dimension that is not within the specified tolerance shall be classified a defect.

4.3.3.1.3 Inspection level and acceptable quality levels (AQL's). The inspection levels and the acceptable quality levels expressed in defects per hundred units shall be as follows:

	Inspection Level	AQL Major	AQL Total
For defects applicable to 4.3.3.1.1	II	2.5	10.0
For defects applicable to 4.3.3.1.2	S-4	(one class)	2.5

4.3.4 Examination of preparation for delivery requirement. An examination shall be made to determine that packaging, packing, and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Shipping containers fully prepared for delivery shall be examined for the defects of closure listed below. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location Sequence or method of application.
Materials	Any component missing, any component damage, affecting Serviceability.
Workmanship	Inadequate application of components such as incomplete Closure of container flaps, inadequate stapling.
Contents	Number per container is more or less than required.

4.4 TESTS. Testing shall be in accordance with FED-STD-191. Fastness shall be in accordance with method 5804.

5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense

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Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The Red Cross panel marker covered by this specification is intended to be used on teenage to designate that the tent is being used for medical purpose.

6.2 Acquisition requirements. Procurement documents should specify the following:

(a) Title, date and number of this specification.
(b) When a standard sample will not be furnished (see 3.1).
(c) When first article is not required (see 3.2).
(d) Selection of applicable levels of packaging and packing (see 5.1)

6.3 Subject term (key word) listing.

Cloth laminated, vinyl nylon
Construction
Grommet
Marking
Materials
Repairs
Slips, tent
Stitching
Tent lines
Thread, polyester
Workmanship

6.4 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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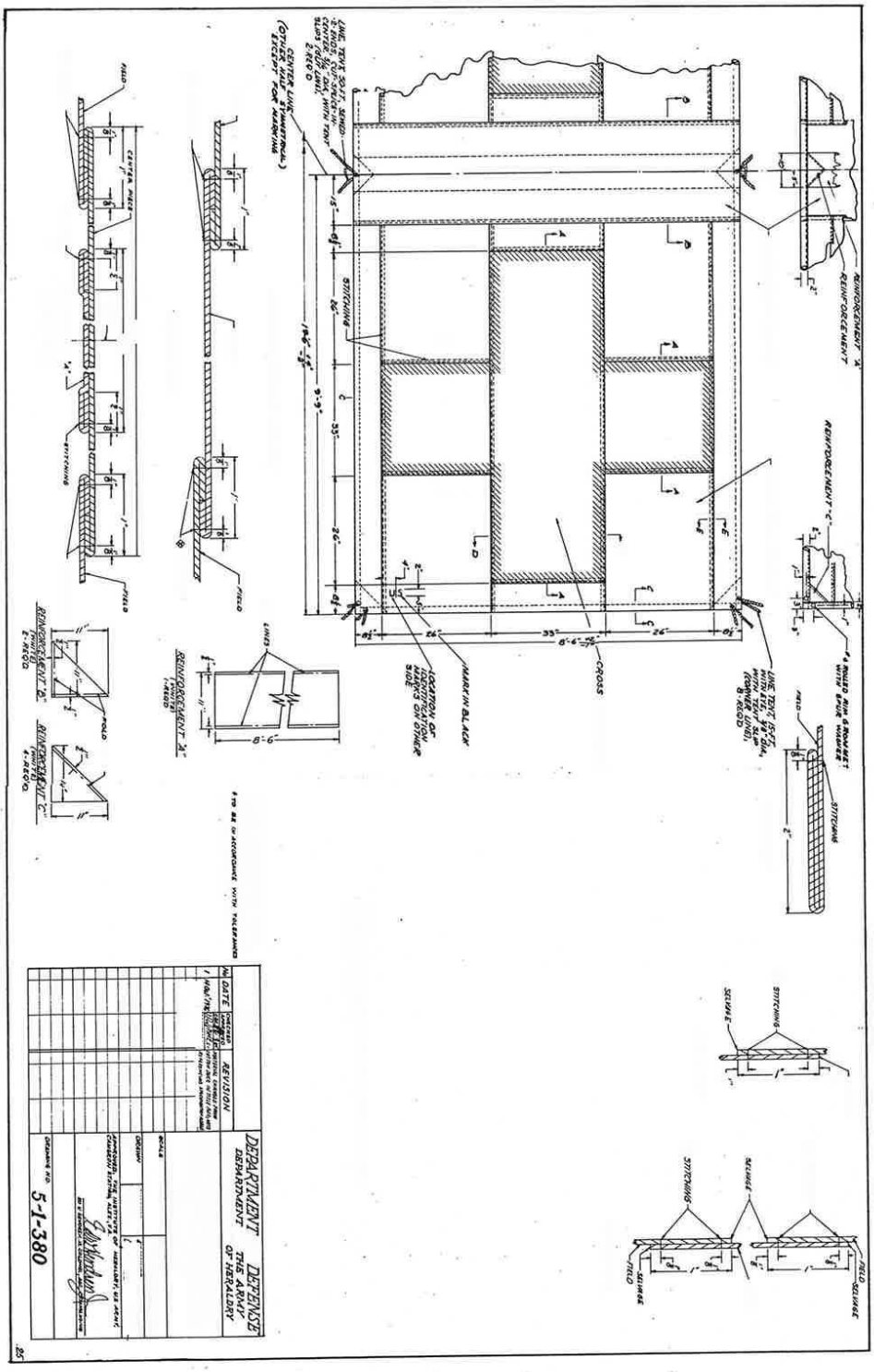


Figure 1: TIOH 5-1-380 Panel Marker, Red Cross

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Army - IH
Navy - NU
Air Force - 11

Preparing Activity:
Army - IH

Review Activity:
Navy - MC
DLA - CT

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